Project Plan

IBBDiS Feedback & Display Interface

Internship – Applied Computer Science

Thomas More University of Applied Sciences
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Duration: 14 weeks

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1. Introduction

This document is the project plan of the internship that was made in the European Commission, in the DG SCIC (Directorate-General for Interpretation) department, as part of a course of the bachelor program at Thomas More. The internship takes place a 14-week period from February 23rd, 2026 to May 22nd, 2026.

The project is about extending the existing IBBDiS (Internal Browser Based Digital Signage) platform with an interactive feedback collection system. IBBDiS is a centralized platform that displays meeting information on digital screens across EC buildings, currently serving more than 1,400 meeting rooms. The system needs to provide live room status information, such as indicating whether a camera is on, and displays the following meeting information on the meeting room panels: current time and date, room name, ongoing meeting name with scheduled time, and upcoming meeting name with scheduled time. At present, there is no mechanism for collecting feedback from meeting participants.

The goal of this document is to describe the background, objectives, scope, planning, progress reporting structure, and team responsibilities of the internship project.

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2. Company Background

2.1 European Commission – DG SCIC

The European Commission is the main executive body of the EU, responsible for proposing legislation, and managing EU decisions. DG SCIC is the department that handles that handles the interpretation services for EU institutions and takes care all the conferences and meeting infrastructure across EC buildings in Brussels.

DG SCIC manages a large number of meeting rooms and conference facilities, handling approximately 1,400 meeting rooms through its internal systems. To support this, it relies on a digital signage infrastructure that communicates real-time meeting information to staff and visitors.

2.2 IBBDiS – Internal Browser Based Digital Signage

IBBDiS is a centralized platform developed and maintained by DG SCIC to display meeting information on digital screens across EC buildings. It pulls meeting data from multiple internal EC systems and shows it on screens installed outside meeting rooms and at building entrances.

The screens currently run on Raspberry Pi devices and display the following information on the meeting room panels:

- Current time and date
- Room name
- Ongoing meeting name and scheduled time
- Upcoming meeting name and scheduled time

Larger screens at building entrances and floor levels use a separate datagrid template showing the full schedule across multiple rooms. The system also has TV-sized screens in some locations that run the same templates but are not interactive.

Meeting Room Categories

The EC meeting rooms are categorized into three types:

Type A — The smallest meeting rooms, used for smaller internal gatherings.

Type B — Medium-sized rooms equipped with sophisticated camera and audio.

Type C — The largest rooms, designed for major meetings and conferences. These rooms have interpreters present inside and are currently equipped with the legacy AMX screens that are being phased out and replaced with the new MTR panels as part of this project.

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3. Current Situation

The current IBBDiS system operates entirely in one direction: it displays information to people but does not collect any input from them. The meeting room panels currently show a two-state display: the ongoing meeting and the upcoming meeting.

The Type A and Type B rooms already have the new Chipsee touchscreen panels installed outside the rooms. The Type C rooms still have the legacy AMX screens inside the room, which are being removed and replaced.

Currently there is no feedback system at all. DG SCIC tracks meeting satisfaction and currently operates at an average score of 4.3 out of 5, which represents a 90% satisfaction rate. The target would be nice to reach 95%. DG SCIC wants to be able to monitor the meeting room experience at a detailed level. So, if the overall feedback for a room on full day is negative, the team can track it and look into the reason by asking the people the used the meeting room. However, if for the whole day received just one or two negative feedback while the rest were positive, that means that it was most likely something not important. Without this kind of system it is not possible to tell the difference, or know where things actually need to be fixed.

4. Project Objectives

The main objectives of this internship project that must be completed and some additional ideas that would be nice to have if time allows are:

- **Collect feedback via Smiley System:** Design and implement an interactive satisfaction survey on the Chipsee touchscreen panels for meeting participants to rate their experience of the meeting room using a simple smiley-face scale from 1 to 5.
- **A TV size Template:** Create a version of the meeting room template for TV sized screens that have not touch, making the same information displayed without interaction capabilities.
- **Redesigned Display Interface:** Extend the existing two-state display (ongoing and upcoming meeting) by adding a third state showing the previous meeting, resulting in with three states: previous, ongoing, and upcoming, in a clear and readable design that follows natural reading patterns and EC visual standards.
- **Redesigned Datagrid Template:** Redesign the large-screen datagrid template that are displayed at building entrances and floors by improving it visually.
- **Inactivity and Burn-in Protection:** Create solutions to handle screen inactivity and prevent burn-in damage on the panels. After analysing the Chipsee, it was clear that they are built for continuous 24/7 and do not require burn in protection.
- **Accessibility Compliance:** Making sure all templates work for everyone, including for participants with color vision deficiency and following the WCAG guidelines.
- **Room Status Indicator:** Design an indicator that shows on the screen whenever it detects the presence of a person or not.
- **Documentation:** Create a detailed technical documentation explaining everything made during this internship.

Nice to Have:

- **EC Survey Integration on trolley Screens:** Add the EU Survey platform on a screen mounted on trolley that can be moved around the building. This way it would allow participants to submit more detailed feedbacks.
- **Dual Feedback System:** Create two feedback types, the quick smiley rating on the Chipsee panels outside the room and a detailed EC Survey on the trolley screen.
- **Proximity Sensor for Screen Wake:** Add a proximity sensor so the screen activates automatically when someone approaches, to make sure it saves power.

- **Android Application:** It would be nice to implement it in a system Android to take advantage of the Android hardware.

5. Business Case

This project delivers several concrete benefits for DG SCIC and the European Commission.

Improved Satisfaction Tracking

The current satisfaction average is 4.3/5, but is only available in the bigger meeting spaces. Satisfaction of smaller spaces cannot be monitored reliably as there is no feedback system. The feedback system will provide accurate, real-time data that allows DG SCIC to track progress and identify problems.

Replacement of Legacy Hardware

The Chipsee panels represent a significant infrastructure investment (approximately €452 per screen plus €600 installation). This project ensures that investment is fully utilized by adding interactive capabilities on top of what the current system already does.

Better Meeting Room Experience

A better visual display interface helps staff and visitors quickly understand the room schedule at a glance. Showing previous, ongoing, and upcoming meetings reduces confusion and improves the flow of people in and out of meeting spaces.

Dual Feedback Channels

By combining two complementary feedback methods, a quick smiley rating on the Chipsee panels for immediate responses and a detailed EC Survey on trolley screens for more in-depth feedback DG SCIC gains a flexible system that caters to both participants who only have a few seconds and those willing to engage more.

Accessibility and Compliance

Ensuring the templates comply with WCAG and support color vision deficiency directly aligns with EU institutional values and legal requirements around accessibility in public-facing digital systems.

Operational Visibility

The room status indicator color provides immediate information about whether a room is in use, which solves the issue that the staff needs check in booking systems manually and prevent accidental interruptions during meetings.

6. Technical Environment

The project operates within the existing IBBDiS infrastructure. The following hardware and software components are relevant:

Hardware

- Chipsee 10-inch touchscreen panels powered by Raspberry Pi, mounted outside meeting rooms (Type A and Type B rooms)
- TV-sized non-interactive screens at various locations

Backend

- Go (programming language used for the IBBDiS backend)
- Internal EC systems: CORALIN, MIRA, EWS, EULearn (meeting data sources)
- PostgreSQL and Oracle databases
- Horizon middleware and Panda API for data flow

Frontend

- HTML templates rendered on devices.
- JavaScript and jQuery for display logic
- EC Square Sans Pro as the primary typeface

Survey & Feedback

- EU Survey platform (ec.europa.eu/eusurvey)

Communication & Collaboration

- Microsoft Teams for team communication and feedback collection
- EC internal documentation standards for all deliverables

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7. Project Scope

7.1 In Scope

- Redesign of the meeting room touchscreen panel template (three-state display: previous, ongoing, upcoming)
- Display of live room information such as camera status, audio, and other meeting indicators
- Backend development to store and process the feedback data collected through the touchscreen panels
- Implementation of an interactive smiley-face feedback survey on the touchscreen panels
- Redesign of the large-screen datagrid template for building entrances and floors
- Creation of a non-interactive TV template for non-touchscreen displays
- Room occupancy status indicator design (red/green visual signal)
- Inactivity and burn-in protection for the panels
- Accessibility review and WCAG compliance of all templates

- Technical documentation covering hardware, infrastructure, design decisions, and version history

7.2 Nice to Have

- Integration of the EU Survey platform on trolley screens for detailed structured feedback
- Proximity sensor integration for automatic screen wake when someone approaches

7.3 Out of Scope

- Physical installation or hardware procurement
- Development of a full analytics dashboard for feedback results

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8. Stakeholders

Stakeholder	Role	Involvement
Internship Mentor	Primary supervisor	Daily guidance, feedback on prototypes, validation of decisions
DG SCIC – IT Colleagues	Technical staff responsible for panels, infrastructure, and hardware	Provide access to panels, load templates, answer technical questions, feedback on prototypes
DG SCIC – AV/Conference Colleague	Manages cameras, projectors, monitors, and conference room equipment	Can assist with hardware integration questions, feedback on prototypes
Administrative Staff	Non-technical staff supporting the internship	Help with access, procedures, and connecting with the right people
Meeting Participants	End users of the touchscreen panels and feedback system	Indirect stakeholders whose experience drives all design decisions
University Supervisor	Academic institution supervising the internship	Evaluates project plan, daily logs, and final report

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9. Definition of Done

A deliverable is considered complete when all of the following conditions are met:

- The template or feature has been tested in a real browser and on the physical Chipsee panel
- The design has been reviewed and approved by the internship mentor

- The template is accessible and has been checked for color vision deficiency compatibility
- The deliverable has been shared with the team via Microsoft Teams and in person at the workplace with no outstanding objections remain
- Everything is fully documented, including configuration, behavioural logic, design decisions, and the reasoning behind what was changed and what was kept

10. Risk Analysis

Risk	Severity	Mitigation
Conflicting feedback from multiple stakeholders leading to unclear design direction	Medium	Consolidate feedback in writing after each round; present options visually and ask stakeholders to vote; defer to mentor for final decisions
Physical panel not available for testing, making it difficult to validate real-world appearance	Medium	Colleague has provided a panel for personal use; new templates can be loaded on request
Survey interaction too complex, resulting in low response rates	High	Keep interaction to a single tap on a smiley face; avoid multi-step flows; test with real users when possible
Templates not displaying correctly on non-touchscreen TV-sized screens	Low	Create a separate non-interactive version of the template early; test on available hardware
Security concerns around data collection and password management	High	Follow EC security guidelines; use EU Survey for data collection; document all libraries used and how credentials are handled
Burn-in damage on panels due to static content	Low	No official solution has been decided yet. The colleague responsible for managing the panels mentioned that burn-in has never been observed on the current panels. After investigation, Burn-in protection was not required.
Proximity sensor implementation not feasible	Low	Colleague confirmed sensors were already tested with minimal impact;
Accessibility issues discovered late in development	Medium	Run color blindness simulation tools on all templates from early iterations; check WCAG compliance at each design round
API access to EC internal network not granted	High	If access is not available, build a mock data system to demonstrate the integration and document it.
Stakeholder not informed about the project, causing work to be paused	Medium	Involve all relevant stakeholders from the start of the project.

11. Project Planning

The internship is 14 weeks, it starts on February 23rd, 2026 and ends May 22nd, 2026. The project progresses through the following phases:

Phase 1 – Onboarding and System Understanding and First Prototype

Weeks 1–2

- Introduction to DG SCIC, the team, and the IBBDiS platform
- Understanding the existing system architecture, data flow, and hardware
- Reviewing existing HTML templates and documentation provided by colleagues
- Beginning HTML prototype development from the first days of the internship
- Iterating on the prototype based on feedback received from colleagues and mentor
- Exploring multiple layout options and running polls to determine the preferred design direction
- Accessibility review using color blindness simulation tools
- Testing templates on the physical Chipsee panel
- Gaining access to the physical panel for local testing
- Sharing prototypes with the team via Microsoft Teams to collect feedback

• Interface Refinement and Additional Templates

Weeks 3–6

- Build the first HTML prototype extending the timetable with a feedback survey
- Share prototype with the team via Microsoft Teams to collect feedback
- Iterate on the design based on stakeholder responses
- Explore multiple layout options and run polls to determine preferred direction
- Begin accessibility review using color blindness simulation tools
- Implementing the final approved template design with three-state display (previous, ongoing, upcoming)
- Simplifying the feedback interaction to a single-tap smiley system
- Designing and implementing the room occupancy status indicator (red/green)
- Beginning work on the non-interactive TV template
- Beginning redesign of the datagrid template for large entrance and floor screens

Phase 3 – Backend Development

Weeks 7–9

- Developing the backend to store and process feedback data collected through the panels Simplify the feedback interaction to a single-tap smiley system
- Design and implement the room occupancy status indicator (red/green)
- Test all templates on the physical panel

Phase 4 – Datagrid Template and Additional Deliverables

Weeks 10–12

- Running full accessibility checks on all templates (WCAG compliance, color vision deficiency)
- Validating all templates on physical hardware
- Collecting final round of feedback from stakeholders
- Addressing remaining issues and finalizing all deliverables
- If the core Chipsee smiley feedback system is fully working and time allows, begin exploring the implementation of the EU Survey platform on i3-Connect trolley screens
- Research on which hardware and what would be needed to deploy EC Survey on it
- Design the interface for the trolley screen to display the structured EC Survey questions
- This phase is optional and will only be started once the Chipsee feedback system is officially complete and approved

Phase 5 – Documentation and Final Report

Weeks 13–14

- Completing all technical documentation including hardware specifications, infrastructure, design decision log, and reasoning behind all design choices
- Writing the final internship report
- Presenting findings and deliverables to the team

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12. Progress Reporting

Regular communication is maintained throughout the project to keep both the workplace mentor and the university supervisor informed.

Daily Logs

A daily internship log is written at the end of each working day and submitted to the supervisor each week. Each entry describes the tasks completed, challenges encountered, and plans for the following day.

Microsoft Teams

Microsoft Teams is the primary communication channel with the DG SCIC team. Prototypes, layout proposals, and questions are shared in the team group chat to collect feedback from all stakeholders.

In-Person Meetings

Regular in-person check-ins with the internship mentor are held to review progress, discuss design decisions, and align on the next steps. Additional meetings are scheduled when specific technical questions or design reviews require it.

Project Management

Progress is tracked through the daily log structure and the iterative feedback cycles conducted via Microsoft Teams. Design versions are documented with notes on what changed between each iteration and why.